

Numerical Analysis chapter 5 theoretical homework

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I. Give a detailed proof of Theorem 5.7.

As we know,

$$\langle u, v \rangle = \int_a^b \rho(t) u(t) \overline{v(t)} dt,$$

where $\rho(x) \in L[a, b]$ is a weight function and $\rho(x) > 0, \forall x$ denotes

$$\|u\|_2 = \left(\int_a^b \rho(t) |u(t)|^2 dt \right)^{\frac{1}{2}}$$

Prove $\mathcal{C}[a, b]$ is an inner-product space over $\mathbb{C}$.

1. $\langle u, u \rangle = \int_a^b \rho(t) u(t) \overline{u(t)} dt = \int_a^b \rho(t) |u(t)|^2 dt > 0$
2. $\langle u, u \rangle = 0 \Leftrightarrow \int_a^b \rho(t) |u(t)|^2 dt = 0 \Leftrightarrow u(t) = 0$ a.e.
3. $\langle u + v, w \rangle = \int_a^b \rho(t) (u(t) + v(t)) \overline{w(t)} dt = \int_a^b \rho(t) u(t) \overline{w(t)} dt + \int_a^b \rho(t) v(t) \overline{w(t)} dt = \langle u, w \rangle + \langle v, w \rangle$
4. $\langle cu, v \rangle = \int_a^b \rho(t) cu(t) \overline{v(t)} dt = c \int_a^b \rho(t) u(t) \overline{v(t)} dt = c \langle u, v \rangle$
5. $\overline{\langle u, v \rangle} = \overline{\int_a^b \rho(t) u(t) \overline{v(t)} dt} = \int_a^b \rho(t) \overline{u(t)} v(t) dt = \langle v, u \rangle$

The proof is completed.

Prove $\mathcal{C}[a, b]$ is a normed space over $\mathbb{C}$.

We know that $\|u\|_2 = \left(\int_a^b \rho(t) |u(t)|^2 dt \right)^{\frac{1}{2}}$.

1. It is obvious that functions in $\mathcal{C}[a, b]$ are commutative and associative.
2. $g = \alpha f \Leftrightarrow g = \alpha f(x), \forall x$
3. Assume that $f_0 = 0$, a.e. $f_0(x) = 0, \forall x, f + f_0 = f$
4. $g(x) = -f(x), g + f = f_0$. We know that f is continuous. $|g(x_1) - g(x_2)| = |f(x_1) - f(x_2)|, \forall x_1, x_2 \in [a, b]. \therefore f$ is continuous and converges, $\therefore g$ is continuous.
5. We have $1 \in \mathbb{R}, \forall f \in \mathcal{C}[a, b], \forall x \in [a, b], 1 * f(x) = f(x)$

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6. By the laws of multiplication and addition, $\mathcal{C}[a, b]$ satisfies distributive laws.

The proof is completed.

II. Consider the Chbyshev polynomials of the first kind.

$$T_n(x) = \cos(n * \arccos(x))$$

a

$$T_n(\cos\theta) = \cos(n\theta)$$

$$\begin{aligned} \langle T_n, T_m \rangle &= \int_{-1}^1 \frac{T_n(x)T_m(x)}{\sqrt{1-x^2}} dx \\ &= \int_{-1}^1 \frac{\cos(n \arccos(x)) \cos(m \arccos(x))}{\sqrt{1-x^2}} dx \\ &= \int_{\pi}^0 \frac{\cos(n\theta) \cos(m\theta)}{\sqrt{1-\cos^2(\theta)}} d(\cos \theta) \\ &= \int_0^{\pi} \cos(n\theta) \cos(m\theta) d\theta \\ &= \int_0^{\pi} \frac{1}{2} (\cos[(n+m)\theta] + \cos[(n-m)\theta]) d\theta \\ &= \frac{1}{2} \left(\frac{\sin[(n+m)\theta]}{n+m} + \frac{\sin[(n-m)\theta]}{n-m} \right) \Big|_0^{\pi} \\ &= 0. \end{aligned}$$

i.e. Chbyshev polynomials of the first kind are orthonormal on $[-1, 1]$ with $\rho(x) = \frac{1}{\sqrt{1-x^2}}$.

b

- $T_0(x) = \cos(0) = 1, \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} dx = \arcsin(x) \Big|_{-1}^1 = \pi.$
 - $T_1(x) = \cos(\arccos(x)) = x, \int_{-1}^1 \frac{x^2}{\sqrt{1-x^2}} dx = \int_{\pi}^0 \frac{\cos^2\theta}{\sin\theta} d(\cos\theta) = \frac{\pi}{2}$
 - $T_2(x) = \cos(2 \arccos(x)) = 2x^2 - 1, \int_{-1}^1 \frac{(2x^2-1)^2}{\sqrt{1-x^2}} dx = \int_{\pi}^0 \frac{(2\cos^2\theta-1)^2}{\sin\theta} d(\cos\theta) = \int_0^{\pi} \cos(2\theta)^2 d\theta = \frac{\pi}{2}$
- So, we have $u_1^* = \frac{1}{\sqrt{\pi}}, \sqrt{\frac{2}{\pi}}x, u_3^* = \sqrt{\frac{2}{\pi}}(2x^2 - 1).$

III

a

(a) By conclusion at II, use the Chbyshev polynomials with weight function $\rho(x) = \frac{1}{\sqrt{1-x^2}}$.

$$u_1^*(x) = \frac{1}{\sqrt{\pi}}, \quad u_2^*(x) = \sqrt{\frac{2}{\pi}}x, \quad u_3^*(x) = \sqrt{\frac{2}{\pi}}(2x^2 - 1).$$

Calculate the Fourier coefficients of $y(x) = \sqrt{1-x^2}$, we have

$$\begin{aligned} b_1 = \langle y, u_0^* \rangle &= \int_{-1}^1 \frac{\sqrt{1-x^2}}{\sqrt{\pi(1-x^2)}} dx = \int_{-1}^1 \frac{1}{\sqrt{\pi}} dx = \frac{2}{\sqrt{\pi}}, \\ b_2 = \langle y, u_1^* \rangle &= \int_{-1}^1 \frac{\sqrt{1-x^2}}{\sqrt{1-x^2}} \sqrt{\frac{2}{\pi}} x dx = \int_{-1}^1 \sqrt{\frac{2}{\pi}} x dx = 0, \\ b_3 = \langle y, u_2^* \rangle &= \int_{-1}^1 \frac{\sqrt{1-x^2}}{\sqrt{1-x^2}} \sqrt{\frac{2}{\pi}} (2x^2 - 1) dx = \int_{-1}^1 \sqrt{\frac{2}{\pi}} (2x^2 - 1) dx = -\frac{2}{3} \sqrt{\frac{2}{\pi}}. \end{aligned}$$

Then the minimizing polynomial of $y(x)$ is

$$\hat{\varphi}(x) = \frac{2}{\sqrt{\pi}} \cdot \frac{1}{\sqrt{\pi}} - \frac{2}{3} \sqrt{\frac{2}{\pi}} \cdot \sqrt{\frac{2}{\pi}} (2x^2 - 1) = -\frac{8}{3\pi} x^2 + \frac{10}{3\pi}.$$

b

Denote the best approximation $\hat{\varphi}(x) = a_0 + a_1 x + a_2 x^2$. Get the Gram matrix of $\{1, x, x^2\}$,

$$G(1, x, x^2) = \begin{bmatrix} \langle 1, 1 \rangle & \langle 1, x \rangle & \langle 1, x^2 \rangle \\ \langle x, 1 \rangle & \langle x, x \rangle & \langle x, x^2 \rangle \\ \langle x^2, 1 \rangle & \langle x^2, x \rangle & \langle x^2, x^2 \rangle \end{bmatrix} = \begin{bmatrix} \pi & 0 & \frac{\pi}{2} \\ 0 & \frac{\pi}{2} & 0 \\ \frac{\pi}{2} & 0 & \frac{3\pi}{8} \end{bmatrix}$$

Then calculate the vector

$$\mathbf{c} = \begin{bmatrix} \langle \sqrt{1-x^2}, 1 \rangle \\ \langle \sqrt{1-x^2}, x \rangle \\ \langle \sqrt{1-x^2}, x^2 \rangle \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ \frac{2}{3} \end{bmatrix}$$

By Theorem 5.34, $G(1, x, x^2)^T \mathbf{a} = \mathbf{c}$, which means

$$\begin{bmatrix} \pi & 0 & \frac{\pi}{2} \\ 0 & \frac{\pi}{2} & 0 \\ \frac{\pi}{2} & 0 & \frac{3\pi}{8} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ \frac{2}{3} \end{bmatrix}$$

Solve the normal equations, we can compute that $a_0 = \frac{10}{3\pi}$, $a_1 = 0$, $a_2 = -\frac{8}{3\pi}$.

Hence $\hat{\varphi}(x) = -\frac{8}{3\pi} x^2 + \frac{10}{3\pi}$.

IV

a

By definition, we have $\|u(t), v(t)\| = \sum_{i=1}^{12} u(t_i) v(t_i)$. and $\|u(t)\| = \sqrt{\sum_{i=1}^{12} u^2(t_i)}$. Apply the Gram-Schmidt process to $(1, x, x^2)$:

- $v_1(x) = 1$, $\|v_1\| = \sqrt{12} = 2\sqrt{3}$, $u_1^*(x) = \frac{v_1(x)}{\|v_1\|} = \frac{1}{2\sqrt{3}}$.
- $v_2(x) = x - \langle x, u_1^*(x) \rangle u_1^*(x) = x - \sum_{i=1}^{12} \frac{x_i}{2\sqrt{3}} \cdot \frac{1}{2\sqrt{3}} = x - 6.5$, $\|v_2\| = \sqrt{\sum_{i=1}^{12} (x_i - 6.5)^2} = \sqrt{143}$. $u_2^*(x) = \frac{v_2(x)}{\|v_2\|} = \frac{1}{\sqrt{143}}(x - 6.5)$.
- $v_3(x) = x^2 - \langle x^2, u_1^*(x) \rangle u_1^*(x) - \langle x^2, u_2^*(x) \rangle u_2^*(x) = x^2 - \sum_{i=1}^{12} \frac{x_i^2}{12} - \sum_{i=1}^{12} x_i^2 (\frac{x_i}{\sqrt{143}} - \frac{1}{2} \sqrt{\frac{13}{11}}) \cdot (\frac{x}{\sqrt{143}} - \frac{1}{2} \sqrt{\frac{13}{11}}) = x^2 - 13x + \frac{91}{3}$, $\|v_3\| = \sqrt{\sum_{i=1}^{12} (x_i^2 - 13x_i + \frac{91}{3})^2} = 2\sqrt{335}$. Thus $u_3^*(x) = \frac{v_3(x)}{\|v_3\|} = \frac{1}{2\sqrt{335}}(x^2 - 13x + \frac{91}{3})$.

Then the list $(\frac{1}{2\sqrt{3}}, \frac{1}{\sqrt{143}}(x - 6.5), \frac{1}{2\sqrt{335}}(x^2 - 13x + \frac{91}{3}))$ is orthonormal.

b

$$\phi(\hat{x}) = \sum_{n=1}^3 c_n u_n^* = \sum_{n=1}^3 \langle y, u_n^* \rangle u_n^*(x)$$

- $c_1 = \langle y, u_1^* \rangle = \sum_{i=1}^{12} y(t_i) u_1^*(t_i) = \sum_{i=1}^{12} \frac{y_i}{2\sqrt{3}} = 277\sqrt{3}$.
- $c_2 = \langle y, u_2^* \rangle = \sum_{i=1}^{12} y(t_i) u_2^*(t_i) = \sum_{i=1}^{12} \frac{y_i(x_i - 6.5)}{\sqrt{143}} = \frac{589}{\sqrt{143}}$.
- $c_3 = \langle y, u_3^* \rangle = \sum_{i=1}^{12} y(t_i) u_3^*(t_i) = \sum_{i=1}^{12} \frac{y_i(x_i^2 - 13x_i + \frac{91}{3})}{2\sqrt{335}} = \frac{6034}{\sqrt{335}}$.

$$\begin{aligned} \phi(\hat{x}) &= c_1 u_1^*(x) + c_2 u_2^*(x) + c_3 u_3^*(x) \\ &= 277\sqrt{3} \cdot \frac{1}{2\sqrt{3}} + \frac{589}{\sqrt{143}} \cdot \frac{1}{\sqrt{143}}(x - 6.5) + \frac{6034}{\sqrt{335}} \cdot \frac{1}{2\sqrt{335}}(x^2 - 13x + \frac{91}{3}) \\ &= 384.908 - 112.959x + 9.00597x^2. \end{aligned}$$

c

The orthonormal list of $u_1^*(x), u_2^*(x), u_3^*(x)$ can be reused since they only depends on the N and x_i . But the Fourier coefficients cannot be reused since they also depends on y_i . If we use the orthonormal polynomials, we can restore them as known conditions, then simplify the process of calculation and only need to calculate the Fourier coefficients. Besides, the method avoid calculating G^{-1} , which might has terrible conditional number and get a catastrophic error.

V, Thm 5.66:

We have $A \in \mathbb{F}^{m \times n}$ and $r = \text{rank}(A)$.

1. $\mathbb{F}^n = \text{span}\{u_1, \dots, u_n\}$
2. $\mathbb{F}^m = \text{span}\{v_1, \dots, v_m\}$

$$\begin{cases} A^+ v_j = \frac{1}{\sigma_j} u_j, \forall j = 1, 2 \cdot r, \\ A^+ v_j = 0, \forall j = r + 1, \dots, m \end{cases} \quad \begin{cases} A^+ u_j = \frac{1}{\sigma_j} v_j, \forall i = 1, 2 \cdot r, \\ A^+ u_j = 0, \forall j = r + 1, \dots, n \end{cases}$$

(a)

$$\begin{aligned} AA^+ A u_j &= AA^+ \sigma_j v_j = \sigma_j v_j = A u_j, j \leq r, \\ AA^+ A u_j &= 0, j > r \end{aligned}$$

So, $AA^+ A = A$.

(b)

$$\begin{aligned} A^+ AA^+ v_j &= A^+ AA^+ \sigma_j u_j = \sigma_j u_j = A v_j, j \leq r, \\ A^+ AA^+ v_j &= 0, j > r \end{aligned}$$

So, $A^+ AA^+ = A^+$.

(c)

$$\begin{cases} AA^+ v_j = A \frac{1}{\sigma_j} u_j = v_j, \forall j = 1, 2 \cdot r, \\ AA^+ v_j = 0, \forall j = r + 1, \dots, m \end{cases} \quad \begin{cases} (AA^+)^* v_j = (A^+)^* A^* v_j = (A^+)^* \sigma_j u_j = v_j, \forall i = 1, 2 \cdot r, \\ (AA^+)^* v_j = 0, \forall j = r + 1, \dots, n \end{cases}$$

(d)

$$\begin{cases} A^+ A u_j = A^+ \sigma_j v_j = u_j, \forall j = 1, 2 \cdot r, \\ A^+ A u_j = 0, \forall j = r + 1, \dots, m \end{cases} \quad \begin{cases} (A^+ A)^* u_j = A^* (A^+)^* u_j = (A^+)^* \frac{1}{\sigma_j} v_j = u_j, \forall i = 1, 2 \cdot r, \\ (A^+ A)^* u_j = 0, \forall j = r + 1, \dots, n \end{cases}$$

So, $(A^+ A)^* = A^+ A$ and $(A A^+)^* = A A^+$.

e

Because A has linearly independent columns, we perform Singular Value Decomposition (SVD) on A first. By Singular Value Decomposition, we have $A = U \Sigma V^*$, where $U \in \mathbb{F}^{m \times m}$, $V \in \mathbb{F}^{n \times n}$ are unitary matrices, and $\Sigma \in \mathbb{F}^{m \times n}$ is a diagonal matrix with singular values $\sigma_1, \dots, \sigma_r$ on the diagonal, $m \geq n$.

$$\begin{aligned} A &= U \Sigma V^* \\ A^* &= V \Sigma^* U^* \\ A^* A &= V \Sigma^* U^* U \Sigma V^* = V \Sigma^* \Sigma V^* \\ \Sigma^* \Sigma &= \begin{bmatrix} \sigma_1^2 & & & \\ & \sigma_2^2 & & \\ & & \ddots & \\ & & & \sigma_r^2 \\ 0 & \dots & 0 & \end{bmatrix} \end{aligned}$$

So, $A^* A$ is invertible.

$$\begin{aligned} (A^* A)^{-1} A^* &= V (\Sigma^* \Sigma)^{-1} V^* U^* \\ &= V \Sigma^+ U^* \\ &= A^+ \end{aligned}$$

From the perspective of mapping, we can derive the following equation:

$$\begin{aligned} A^* A A^+ v_j &= A^* A \frac{1}{\sigma_j} u_j = A^* v_j = A v_j \\ A^* A A^+ v_j &= 0, j > r. \end{aligned}$$

So, $A^* A A^+ = A^*$.

f

Because A has linearly independent rows, we perform Singular Value Decomposition (SVD) on A first. By Singular Value Decomposition, we have $A = U \Sigma V^*$, where $U \in \mathbb{F}^{m \times m}$, $V \in \mathbb{F}^{n \times n}$ are unitary matrices, and $\Sigma \in \mathbb{F}^{m \times n}$ is a diagonal matrix with singular values $\sigma_1, \dots, \sigma_r$ on the diagonal, $m \geq n$.

$$\begin{aligned} A &= U \Sigma V^* \\ A^* &= V \Sigma^* U^* \\ A A^* &= V \Sigma^* U^* U \Sigma V^* = V \Sigma^* \Sigma V^* \\ \Sigma^* \Sigma &= \begin{bmatrix} \sigma_1^2 & & 0 & \dots & 0 \\ & \sigma_2^2 & 0 & \dots & 0 \\ & & \ddots & 0 & \dots & 0 \\ & & & \sigma_r^2 & 0 & \dots & 0 \end{bmatrix} \end{aligned}$$

So, AA^* is invertible. From the perspective of mapping, we can derive the following equation:

$$AA^*v_j = A^+A\sigma_j u_j = A^+\sigma_j^2 v_j = A^*v_j$$

$$A^+AA^*v_j = 0, j > r.$$

So, $A^+AA^* = A^*$.

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